XtalPi: Accelerating drug discovery and development to offer more timely and affordable medical treatments

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About XtalPi

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XtalPi is a pharmaceutical technology company driven by AI and automation. Dedicated to revolutionizing drug development through enhancing the speed, scale, innovation and success rate of drug discovery and development, XtalPi provides services to pharmaceutical companies around the world from its offices in China and the US. Constantly exploring the most optimal solutions, XtalPi strives to fulfill the needs of its customers and partners by fully taking advantage of advanced computing technologies.

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Google Cloud results

- Guarantees data security of multi-cloud architecture through default encryption
- Supports nearly zero computing disruption rate with abundant computing resources on Spot VMs
- Lowers computing costs by 10% with

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XtalPi's AI drug development platform is an experiment-based R&D system with high prediction accuracy through the integration of cloud-based supercomputing digital development tools and advanced experiment capabilities. As a leading AI drug development company, XtalPi has established a full set of development processes with a close combination of quantum physics dry labs and advanced wet labs. By overcoming the efficiency bottlenecks of traditional drug development methods, XtalPi continues to reach higher levels of innovation speed and scale in drug discovery and development.

Industries: Healthcare, Life Sciences

Location: US

(https://cloud.google.com/), Cloud GPUs
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Google Cloud (https://cloud.google.com/)

Cloud GPUs (https://cloud.google.com/gpu/)

Spot VMs (https://cloud.google.com/spot-vms)

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Spot VMs to support more cost-effective drug development

Fast and cost-effective drug development is in demand more than ever. Facing global pandemics like COVID-19 and the higher prevalence of chronic diseases brought by higher life expectancy (https://www.un.org/development/desa/pd/sites/www.un.org.development.desa.pd/files/wpp2022_summary_of_results.pdf), people around the world increasingly need timely-developed and affordable medicines to safeguard their health.

XtalPi (https://www.xtalpi.com/en/about/) is dedicated to bringing more effective medicines to patients faster and at lower costs by helping the pharmaceutical industry accelerate drug discovery and development. To develop a new drug, pharmaceutical companies have to go through a series of processes, from designing a new product, and selecting the most promising compounds, to conducting clinical trials. Traditionally, they would spend billions of dollars and years in research and development (R&D) before a new medicine is released into the marketplace.

Founded in 2014 by a group of quantum physicists from the Massachusetts Institute of Technology (MIT), XtalPi has been using new technologies, such as artificial intelligence (AI), quantum physics, and cloud-based supercomputing, to develop AI algorithms that are able to enhance the efficiency, accuracy, and success rate of medicine R&D. The company started its business by providing pharmaceutical companies with an algorithm-driven solution for crystal structure prediction, a method to identify the most stable crystal structure for drug production.

In 2016, XtalPi participated in a global crystal structure prediction contest held by Pfizer and achieved a perfect prediction accuracy rate of 100%, which led to its long-term partnership with Pfizer (https://www.pfizer.com/news/articles/how_quantum_physics_and_ai_is_disrupting_drug_discovery_development). Using traditional methods, the entire calculation process can take anywhere between two months to over half a year. However, XtalPi is able to use its algorithms to reduce the time needed to less than two weeks. Its solutions can also help significantly shorten the time required for drug discovery.

In the pharmaceutical industry, a highly efficient drug development process not only helps save time and costs, but also ensures that drug companies can obtain a patent before their competitors. That is why XtalPi's products have received increasing popularity. As of November 2022, the company has partnered with more than 150 global pharmaceutical companies, including most of the top 20 enterprises in the industry, and supports around 200 drug development projects, notably Paxlovid (https://www.paxlovid.com/), a widely approved oral drug for COVID-19.

To acquire best-of-breed technologies and ensure sufficient computing resources, XtalPi deploys its drug development platform in a multi-cloud environment, and is constantly looking for more ideal solutions. In 2020, the company decided to incorporate Google Cloud (https://cloud.google.com/) into its multi-cloud computing architecture, because it guarantees high-standard data security while offering computing instances with high stability, availability, and low price-performance ratios.

"To ensure that we can always deliver computational results in time while lowering drug development costs as much as possible, we need computing instances with low cost-performance ratios that can meet our fluctuating computing demand to run our algorithms," explains Chen Yongpan, drug development platform director at XtalPi. "Google Cloud perfectly meets our computing requirements. More importantly, it can help us realize better data security protection."

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Ensuring data security in a multi-cloud architecture

Data security is key to drug development, as competition is stiff in the pharmaceutical industry, and companies need to meet strict security requirements set for the healthcare industry. Being trusted with all the drug development-related data that it processes for its customers, XtalPi prioritizes data security and chose to adopt Google Cloud primarily for its high-standard security safeguard measures. Since Google Cloud offers default encryption for all the data transferred through its network, XtalPi can assure its customers that no data is accessible by cyberattackers. All the sensitive business data of XtalPi's customers can therefore be well protected, enabling pharmaceutical companies to freely explore innovations with their data, which in return helps accelerate drug discovery and development.

On top of that, Google Cloud's approach with the open cloud (https://cloud.google.com/open-cloud) gives XtalPi full control over its infrastructure and data, while leveraging a multi-cloud architecture. Having this full data sovereignty, XtalPi is able to adopt necessary measures to meet the regulatory requirements demanded by different customers. At the same time, Google Cloud is ISO/IEC 27001 compliant, which is aligned with the level of data security of XtalPi's internal systems. As a result, the company can offer a unified data protection standard in all its services.

"We understand that data security is one of the primary concerns of our customers, and we've implemented all the measures needed to ensure it," explains Chen. "With the security features of Google Cloud, we're able to achieve the highest standards of data security and compliance in a multi-cloud environment rather effortlessly. Our customers can therefore take full advantage of our services and focus on product development without worrying about sensitive data being leaked."

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Offering richer computing resources with Spot VMs and Cloud GPUs

One of the advantages that XtalPi noticed about using Google Cloud is the increased availability of computing resources, which enables the company to run algorithms more stably and efficiently. In mid-2022, XtalPi started to run its algorithms for crystal structure prediction on Spot VMs (https://cloud.google.com/spot-vms). Before migrating its algorithm computing workloads to Spot VMs, XtalPi often used up all the computing resources available during peak times. When the computing resources fell short, it would take a much longer time for the XtalPi team to deliver computational results to its customers, because the algorithm running was blocked, and queueing was required. According to Chen, the abundant computing resources provided by Spot VMs translate into nearly zero disruption and very low queuing time, which enables the XtalPi team to deliver computational results on time.

Additionally, XtalPi employs Cloud GPUs (https://cloud.google.com/gpu) to conduct most of its

GPU computing for new drug development. Since Cloud GPUs offer more than 10,000 GPUs, in contrast with a few thousands of GPUs provided by other similar offerings, the XtalPi team is able to guarantee an 8-hour result delivery for all its GPU computing tasks.

"Having abundant computing resources at our disposal is crucial for us, because our business has been growing rapidly, and our customers might ask us to redo computing tasks any time. With the help of Cloud GPUs, we're able to assure all our customers that we can deliver results in a short time, no matter how big the computing tasks are," says Chen.

Reducing computing costs while enhancing network connection speed

Before leveraging Spot VMs, XtalPi conducted a new round of benchmark tests to identify the virtual machines (VMs) that could best meet its computing needs. After comparing different offerings, the XtalPi team found that Spot VMs of Google Cloud have lower price-performance ratio, and support network connection speeds that are four times faster than the second product in the ranking.

Chen says that XtalPi always strives to minimize drug development costs to bring more affordable medicines to the market, and leveraging Spot VMs can help the company pursue this goal by lowering the computing costs by 10%. At the same time, the great connectivity supported by the extensive global network of Google Cloud means that Xtalpi can run the computing tasks involving frequent data transferring more efficiently.

"We're constantly enhancing the operational efficiency of our algorithms by choosing the most ideal VMs to deploy our workloads," notes Chen. "Spot VMs are currently a better solution that can help us reduce drug development costs while enhancing the quality of our service."

Expanding service range with high-performance computing instances

Moving forward, XtalPi plans to bring its AI-powered solutions to other industries, including agricultural and material science. As the company expands its product lines, its demand for high-performance computing instances will increase significantly. Chen says that the cost-effective computing resources of Google Cloud, like Spot VMs, will play an important role in the execution of the company's business expansion plan.

He adds, "Google Cloud has helped us realize highly efficient drug development and discovery by providing high-performance and abundant computing resources at lower costs. We look forward to seeing how it will continue facilitating our work to revolutionize the pharmaceutical industry with AI, and expand the application of our technology."

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